**Department of Computer Science and Engineering**

**CSA10 – CO2 AT1**

Software Engineering

# Scenario-Based Requirement Analysis

*CampusBite: Smart Campus Food Ordering System*

| Detail | Information |
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# 1. Introduction

Requirement analysis is the foundation of every software project. Errors made at this stage are the most expensive to correct later, so the needs of all stakeholders must be captured, analysed, prioritised and validated before design begins. This assignment applies the requirement engineering process to a realistic scenario: a campus food ordering system called CampusBite.

## 1.1 Objectives

- Identify the stakeholders and their needs in the given scenario.
- Elicit, document and classify functional and non-functional requirements.
- Model key interactions through user stories and a detailed use case.
- Prioritise and validate the requirements and trace them to business goals.

# 2. Scenario Description

A large engineering college has about 6,000 students and 400 staff members spread across several blocks. The campus has three canteens and a coffee kiosk. During the lunch break of 40 minutes, queues at the counters often exceed 15 minutes, many students skip meals or arrive late to class, food is wasted because canteens cannot predict demand, and cash handling causes billing errors and delays.

The college management has decided to build CampusBite, a web and mobile platform that lets students and staff browse live menus, pre-order food, pay online and collect it at a scheduled pickup slot. Canteen managers will use the same platform to manage menus, stock and orders, and the administration will use it to monitor sales and wastage.

## 2.1 Problem Statement

The existing manual counter-based process is slow, error-prone and unable to scale during peak hours. A digital ordering system is required that reduces waiting time, improves billing accuracy, supports cashless payment, and gives management data for planning.

## 2.2 Business Goals

- Reduce average waiting time per order from about 15 minutes to under 3 minutes.
- Reduce food wastage by at least 20% through demand-based preparation.
- Achieve at least 70% cashless transactions within one semester of launch.
- Provide daily and monthly sales reports without manual compilation.

## 3. Stakeholder Analysis

| Stakeholder | Role | Key Interests / Needs | Influence |
|---|---|---|---|
| Students | Primary user | Fast ordering, accurate menu, easy payment, pickup without queues | Medium |
| Faculty and staff | Primary user | Quick ordering between classes, order history, receipts | Medium |
| Canteen managers | Primary user | Manage menu and stock, view incoming orders, reduce wastage | High |
| Kitchen staff | Secondary user | Clear order queue, preparation status updates | Low |
| College administration | Sponsor | Cost control, sales reports, student satisfaction | High |
| IT department | Maintainer | Easy deployment, security, integration with campus login | High |
| Payment provider | External party | Secure, standards-compliant transaction handling | Medium |
| Parents / Management board | Indirect | Hygiene, healthy food options, transparency | Low |

## 4. Requirement Elicitation

A combination of techniques was chosen so that requirements are gathered from all stakeholder groups and cross-checked.

| Technique | Applied To | Outcome |
|---|---|---|
| Interviews | Canteen managers, administration, IT staff | Business rules, stock handling process, reporting needs |
| Questionnaires / surveys | Students and faculty (sample of 300) | Preferred payment modes, peak hours, required features |
| Observation | Canteen counters during lunch break | Queue patterns, billing bottlenecks, order mix |
| Document analysis | Existing bills, stock registers, complaint records | Data fields, frequent errors, wastage figures |
| Prototyping | Paper and wireframe screens | Early feedback on ordering flow and menu layout |
| Brainstorming workshop | Cross-functional group | Additional features such as pre-scheduling and feedback |

## 5. Functional Requirements

Functional requirements describe what the system must do. Priority follows the MoSCoW scheme (M = Must have, S = Should have, C = Could have).

| ID | Requirement | Actor | Priority |
|---|---|---|---|
| FR-01 | The system shall allow users to register and log in using their college ID and password or campus single sign-on. | All users | M |
| FR-02 | The system shall display the live menu of each canteen with price, availability, ingredients and veg / non-veg tags. | Student, Staff | M |
| FR-03 | The system shall allow users to search and filter menu items by category, price and dietary type. | Student, Staff | S |
| FR-04 | The system shall allow users to add items to a cart, modify quantities and place an order. | Student, Staff | M |
| FR-05 | The system shall allow users to choose a pickup time slot and reject slots that are full. | Student, Staff | M |

| ID | Requirement | Actor | Priority |
|---|---|---|---|
| FR-06 | The system shall support online payment through UPI, cards and campus wallet, and generate a digital receipt. | Student, Staff | M |
| FR-07 | The system shall notify the user when the order is accepted, ready for pickup, or cancelled. | Student, Staff | M |
| FR-08 | The system shall allow users to cancel an order before preparation starts and refund the amount automatically. | Student, Staff | S |
| FR-09 | The system shall show the kitchen a real-time queue of orders and allow status updates. | Kitchen staff | M |
| FR-10 | The system shall allow canteen managers to add, edit, disable menu items and update stock. | Canteen manager | M |
| FR-11 | The system shall mark items as 'out of stock' automatically when stock reaches zero. | System | S |
| FR-12 | The system shall generate daily, weekly and monthly sales and wastage reports. | Manager, Admin | M |
| FR-13 | The system shall allow users to view order history and re-order a past order. | Student, Staff | S |
| FR-14 | The system shall allow users to rate items and submit feedback after pickup. | Student, Staff | C |
| FR-15 | The system shall allow administrators to manage users, canteens and role permissions. | Admin | M |
| FR-16 | The system shall suggest popular and previously ordered items to users. | System | C |

## 6. Non-Functional Requirements

| ID | Category | Requirement | Measure |
|---|---|---|---|
| NFR-01 | Performance | Pages shall load quickly and orders shall be confirmed promptly during peak hours. | Page load under 2 s; order confirmation under 3 s at 1,500 concurrent users |
| NFR-02 | Availability | The system shall be available during canteen working hours. | 99.5% uptime between 7 AM and 8 PM |
| NFR-03 | Scalability | The system shall handle growth in users and canteens without redesign. | Support 10,000 users and 10 canteens by horizontal scaling |

| ID | Category | Requirement | Measure |
|---|---|---|---|
| NFR-04 | Security | User data and payments shall be protected. | HTTPS everywhere, passwords hashed, role-based access, PCI-DSS compliant gateway |
| NFR-05 | Usability | The interface shall be simple enough to use without training. | A new user places an order within 2 minutes; WCAG 2.1 AA accessibility |
| NFR-06 | Reliability | No order or payment shall be lost or duplicated. | Zero lost transactions; automatic retry and reconciliation |
| NFR-07 | Compatibility | The system shall run on common devices and browsers. | Android 9+, iOS 14+, latest Chrome, Edge, Safari, Firefox |
| NFR-08 | Maintainability | The system shall be modular and documented. | Code coverage of at least 70%; deployment of a fix within 1 hour |
| NFR-09 | Privacy / Legal | Personal data shall be handled as per applicable data protection law. | Consent recorded; data retained for no more than 2 years |
| NFR-10 | Recoverability | The system shall recover from failures quickly. | RPO 15 minutes; RTO 1 hour; daily backups |

# 7. Constraints and Assumptions

## 7.1 Constraints

- Budget: the project must be delivered within the budget approved by the college administration.
- Schedule: the first release must be live before the start of the next semester (about 4 months).
- Technology: open-source technologies are preferred to avoid licence costs.
- Integration: the system must integrate with the existing campus student database and authentication service.
- Payments: only RBI-approved payment gateways may be used.

## 7.2 Assumptions

- All students and staff own a smartphone with internet access, or can use the web portal on campus Wi-Fi.
- Campus Wi-Fi is stable enough in the canteen areas.
- Canteen staff will receive one day of training before launch.
- Menu and price data will be supplied by the canteen managers.

# 8. User Stories

User stories express requirements from the user's point of view in the format: As a <role>, I want <goal> so that <benefit>.

| ID | User Story | Acceptance Criteria | Linked FR |
|---|---|---|---|
| US-01 | As a student, I want to see the live menu so that I only order items that are available. | Out-of-stock items are greyed out and cannot be added to the cart. | FR-02, FR-11 |
| US-02 | As a student, I want to pre-order and pick a time slot so that I do not stand in a queue. | Only slots with free capacity are selectable; confirmation shows the slot. | FR-04, FR-05 |

| ID | User Story | Acceptance Criteria | Linked FR |
|---|---|---|---|
| US-03 | As a staff member, I want to pay by UPI so that I do not need cash. | Payment succeeds or fails clearly; receipt is generated on success. | FR-06 |
| US-04 | As a kitchen staff member, I want a live order queue so that I can prepare food in the right sequence. | New orders appear within 3 seconds, sorted by pickup time. | FR-09 |
| US-05 | As a canteen manager, I want to update stock and menu so that students see correct information. | Changes appear to users within 1 minute. | FR-10 |
| US-06 | As an administrator, I want sales and wastage reports so that I can plan purchases. | Reports can be filtered by date and canteen and exported to PDF or Excel. | FR-12 |
| US-07 | As a student, I want to cancel an order so that I get a refund if my plans change. | Cancellation is allowed until preparation starts; refund is initiated automatically. | FR-08 |

# 9. Use Case Specification

## 9.1 Use Case Overview

| Use Case ID | Use Case | Primary Actor |
|---|---|---|
| UC-01 | Log in | Student, Staff, Manager, Admin |
| UC-02 | Browse menu | Student, Staff |
| UC-03 | Place pre-order | Student, Staff |
| UC-04 | Make payment | Student, Staff, Payment gateway |
| UC-05 | Prepare and update order status | Kitchen staff |
| UC-06 | Manage menu and stock | Canteen manager |
| UC-07 | Generate reports | Manager, Admin |

### 9.2 Detailed Use Case: UC-03 Place Pre-order

| Field | Description |
|---|---|
| Primary actor | Student or staff member |
| Secondary actors | Payment gateway, Notification service |
| Goal | Order food from a canteen for pickup at a chosen time. |
| Preconditions | User is logged in; selected canteen is open; at least one item is available. |
| Trigger | User selects 'Place Order' from the cart. |
| Main flow | 1. User browses the menu and adds items to the cart.<br>2. System shows cart total and available pickup slots.<br>3. User selects a pickup slot.<br>4. System checks stock and slot capacity.<br>5. User confirms and selects payment mode.<br>6. System sends the payment request to the gateway.<br>7. Gateway confirms payment.<br>8. System stores the order, reduces stock and sends it to the kitchen queue.<br>9. System sends confirmation and receipt to the user. |
| Alternate flows | A1. Item becomes unavailable: system informs the user and removes the item.<br>A2. Selected slot is full: system suggests the nearest free slot. |
| Exception flows | E1. Payment fails: order is not created; user may retry or change payment mode.<br>E2. Network is lost after payment: system reconciles with the gateway and confirms or refunds. |

| Field | Description |
|---|---|
| Postconditions | Order is stored with status 'Accepted'; stock is updated; user is notified. |
| Business rules | Order must be placed at least 10 minutes before the pickup slot; one order per slot can contain at most 10 items. |

## 10. Requirement Prioritisation

Requirements were prioritised using the MoSCoW technique after discussion with the sponsor and canteen managers.

| Category | Meaning | Requirements |
|---|---|---|
| Must have | Release cannot go live without these. | FR-01, FR-02, FR-04, FR-05, FR-06, FR-07, FR-09, FR-10, FR-12, FR-15 |
| Should have | Important, but a work-around exists. | FR-03, FR-08, FR-11, FR-13 |
| Could have | Desirable if time permits. | FR-14, FR-16 |
| Won't have (this release) | Deferred to a later version. | Loyalty points, delivery to hostel rooms, multi-language voice ordering |

## 11. Requirement Validation and Traceability

### 11.1 Validation Techniques

- Requirement reviews and walkthroughs with stakeholders to confirm completeness and correctness.
- Checking each requirement for the SMART qualities: specific, measurable, achievable, relevant and testable.
- Prototype demonstration to canteen managers and a sample group of students.

- Creating test cases from acceptance criteria to ensure each requirement is verifiable.
- Consistency check to find conflicts, for example between the 'order at least 10 minutes before pickup' rule and 'instant ordering' expectations.

## 11.2 Requirements Traceability Matrix (extract)

| Business Goal | Requirements | Planned Test |
|---|---|---|
| Reduce waiting time | FR-04, FR-05, FR-09, NFR-01 | Load test with 1,500 users; timed order flow test |
| Cashless payments | FR-06, FR-08, NFR-04, NFR-06 | Payment success, failure and refund test cases |
| Reduce wastage | FR-10, FR-11, FR-12 | Report accuracy check against stock register |
| Management reporting | FR-12, FR-15 | Report generation and export test |

## 11.3 Conflicts and Risks

| Issue | Resolution |
|---|---|
| Students want last-minute ordering; kitchens need preparation time. | Fixed minimum lead time of 10 minutes with fast-prep items marked. |
| Online payments may fail due to network issues. | Payment reconciliation job and automatic refund (NFR-06). |
| Peak load at 12:30 PM may overload the system. | Caching, queue-based processing and horizontal scaling (see AT2 architecture). |
| Resistance from counter staff to a new process. | Training session and a simple dashboard. |

## 12. Conclusion

This assignment analysed the CampusBite scenario by identifying stakeholders, eliciting needs through multiple techniques and documenting 16 functional and 10 non-functional requirements. The requirements were expressed as user stories and a detailed use case, prioritised using MoSCoW, and validated and traced to business goals. The resulting requirement set is clear, testable and consistent, and forms the input for the architecture design presented in CO2 AT2.